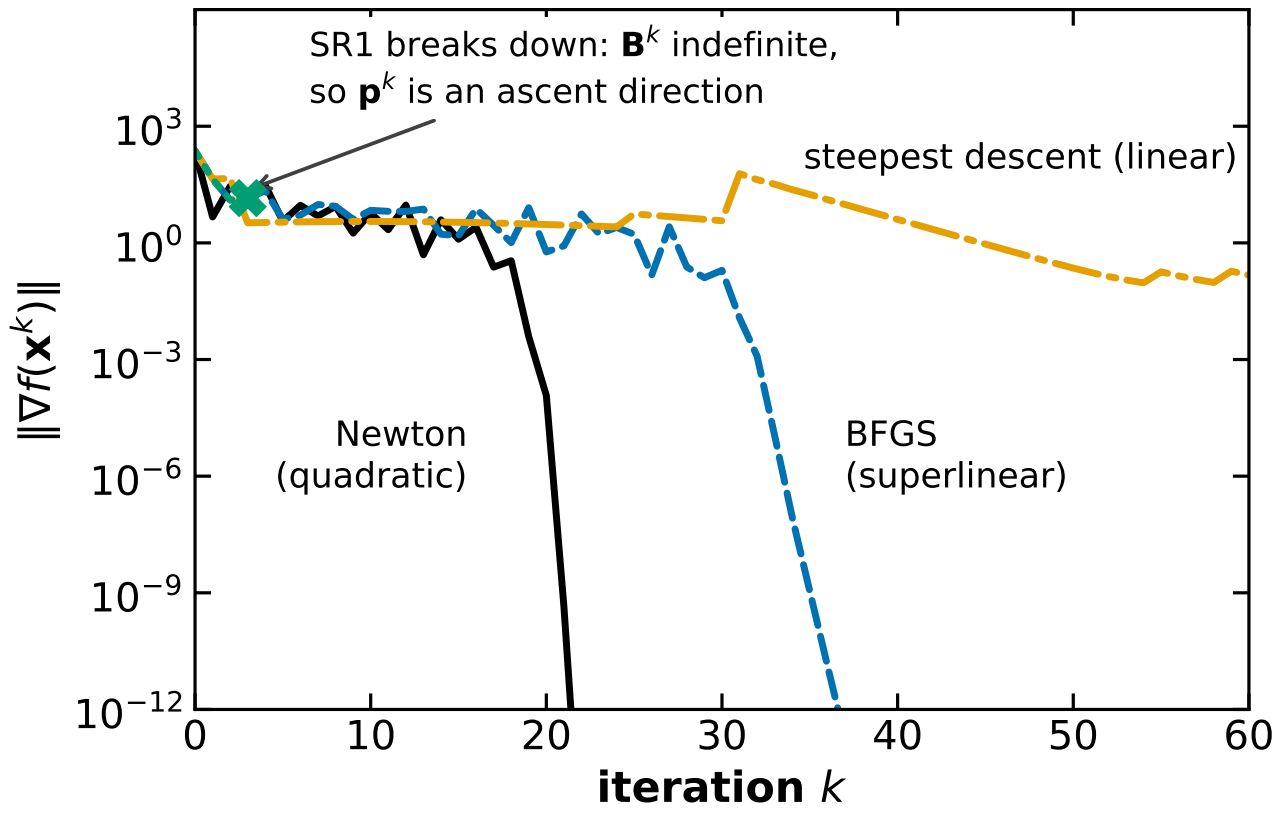
SR1 breaks down: $\mathbf{B}^k$ indefinite,
so $\mathbf{p}^k$ is an ascent direction
steepest descent (linear)
Newton
(quadratic)
BFGS
(superlinear)
$\|\nabla f(\mathbf{x}^k)\|$
10^{3}
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
0
10
20
30
40
50
60
iteration k